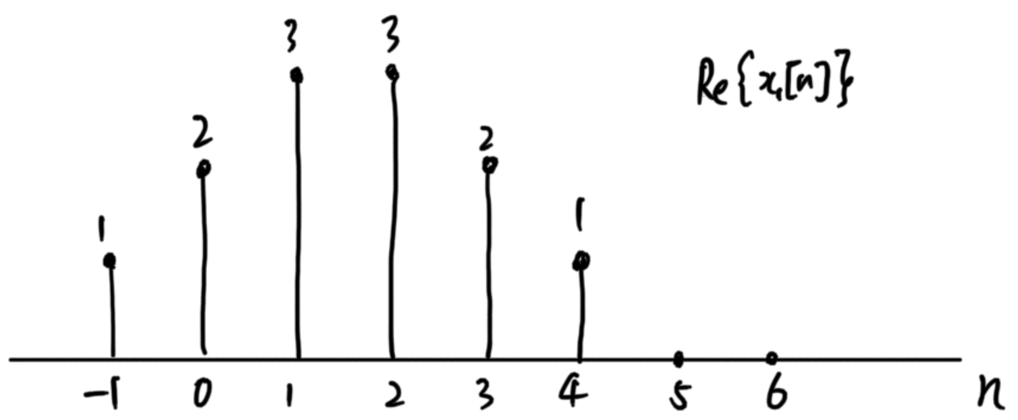
2.44 (a) $X(e^{j\omega})\Big|_{\omega=0} = \sum_{n=-\infty}^{+\infty} x[n] = 12$

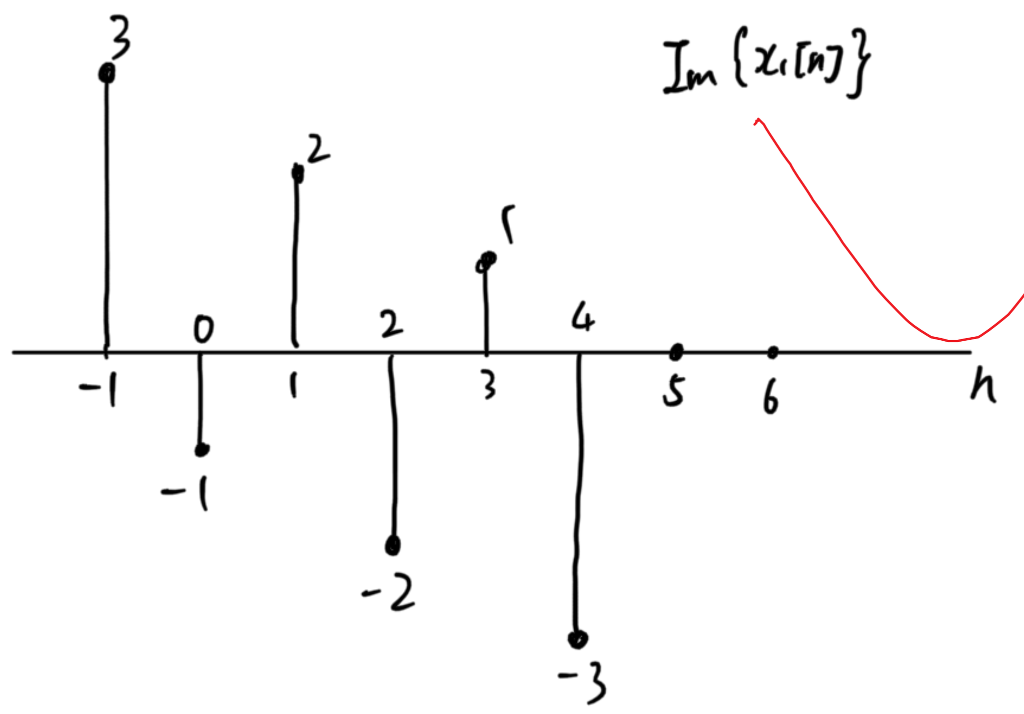
(b) $X(e^{j\omega})\Big|_{\omega=\pi} = \sum_{n=-\infty}^{+\infty} x[n](-1)^n = -12i$

(c) $\int_{-\pi}^{\pi} X(e^{j\omega})\,d\omega = 2\pi x[0] = 4\pi - 2\pi i$

(d) $X(e^{-j\omega}) = \sum_n x[n] e^{j\omega n} = \sum_n x[-n] e^{-j\omega n}$

所以 $x[n] \Leftrightarrow X(e^{j\omega}) \Leftrightarrow$ 所求信號 $x_1[n] = x[-n] \Leftrightarrow X(e^{j\omega})$

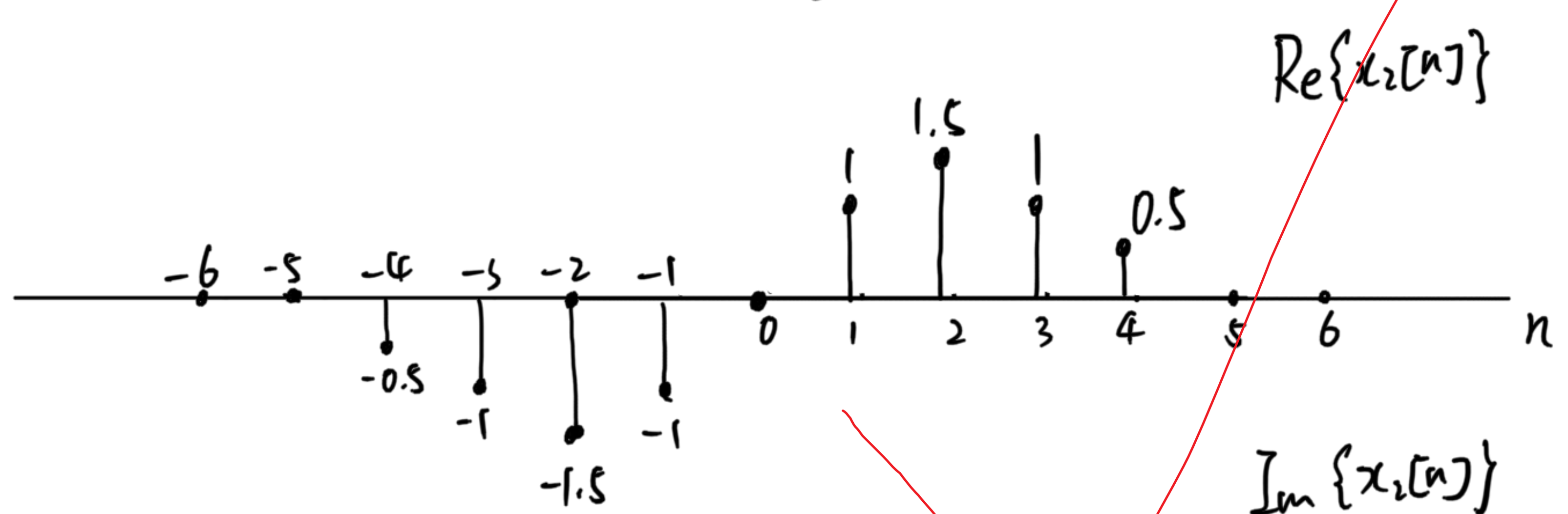


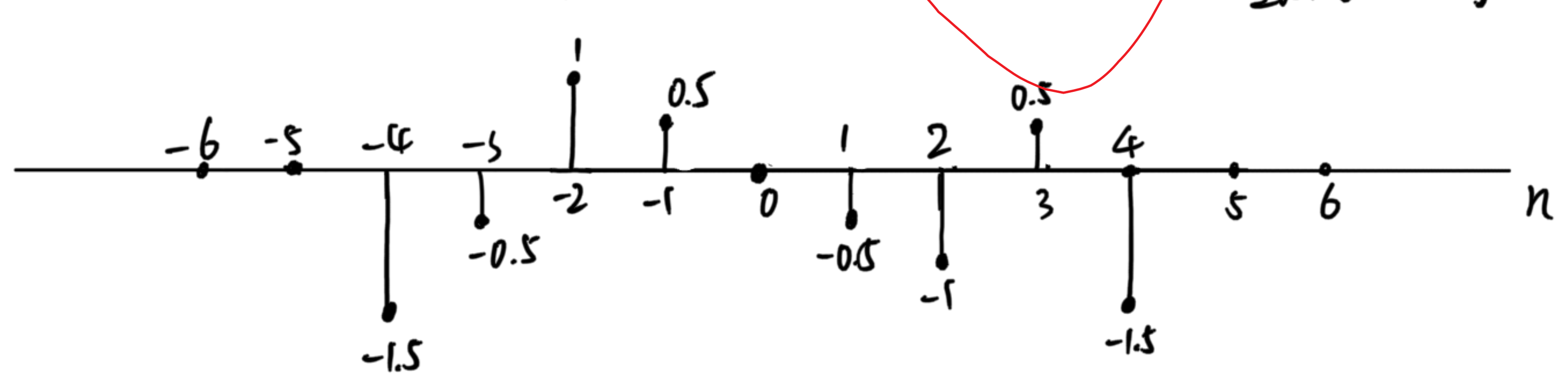


(e) $x[n] = x_o[n] + x_e[n]$, $X_o(e^{j\omega}) = jI_m\{X(e^{j\omega})\}$

所以 $x_o[n] \Leftrightarrow X_o(e^{-j\omega}) = jI_m\{X(e^{j\omega})\}$

所求信號 $x_2[n] = x_o[-n] = \frac{x[-n]-x[n]}{2} = \frac{x_1[n]-x_1[-n]}{2}$





2.89 (a) $\Phi_{xx}[m] = E\left[\sum_{n=-\infty}^{+\infty} x[n]x[n+m]\right] = \Phi_{ss}[0] + \Phi_{ee}[0] + E\left[\sum_{n=-\infty}^{+\infty} s[n]e[n+m]\right] + E\left[\sum_{n=-\infty}^{+\infty} e[n]s[n+m]\right]$

e[n], s[n] 独立
互相关为0

$= \Phi_{ss}[m] + \Phi_{ee}[m] + \Phi_{se}[m] + \Phi_{es}[m]$

$\Phi_{xx}(e^{j\omega}) = \sum_{m=-\infty}^{+\infty} \Phi_{xx}[m] e^{-j\omega m} = (\Phi_{ss}[0]+\Phi_{ee}[0])\cdot 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega-2k\pi) + \Phi_{se}(e^{j\omega}) + \Phi_{es}(e^{j\omega})$

$= (\Phi_{ss}[0]+\Phi_{ee}[0])\cdot 2\pi \sum_{k=-\infty}^{+\infty} \delta(\omega-2k\pi) + 2Re\{\Phi_{es}(e^{j\omega})\}$

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(b) $\Phi_{xe}[m] = E\left[\sum_{n=-\infty}^{+\infty} x[n]e[n+m]\right] = \Phi_{se}[m] + \Phi_{ee}[m]$

$\Phi_{xe}(e^{j\omega}) = \Phi_{se}(e^{j\omega}) + \Phi_{ee}(e^{j\omega})$

(c) $\Phi_{xe}[m] = E\left[\sum_{n=-\infty}^{+\infty} x[n]s[n+m]\right] = \Phi_{ss}[m] + \Phi_{es}[m]$

$\Phi_{xe}(e^{j\omega}) = \Phi_{ss}(e^{j\omega}) + \Phi_{es}(e^{j\omega})$

2.92 (a) $\sigma_x^2 = E[x[n]^2]$

$E[x[n]y[n]] = E\left[x[n]\sum_{m=-\infty}^{+\infty} h[m]x[n-m]\right] = \sum_{m=-\infty}^{+\infty} h[m]R_{xx}[m] = h[0]\sigma_x^2$

(b) $E[y[n]] = E[x[n]]\sum_{k=-\infty}^{+\infty} h[k] = 0$

$\sigma_y^2 = E[y[n]^2] = E\left[\left(\sum_{m=-\infty}^{+\infty} h[m]x[n-m]\right)^2\right] = E\left[\sum_{m=-\infty}^{+\infty}\sum_{k=-\infty}^{+\infty} h[m]h[k]x[n-m]x[n-k]\right]$

$= \sum_{m=-\infty}^{+\infty}\sum_{k=-\infty}^{+\infty} h[m]h[k]R_{xx}[m-k]$

$= \sum_{m=-\infty}^{+\infty} h[m]\,(h[m] * R_{xx}[m])$

$= \sigma_x^2 \sum_{m=-\infty}^{+\infty} h[m]h[m] = \sigma_x^2 \sum_{n=-\infty}^{+\infty} h[n]^2$

2.96 (a) $E[x^2[n]] = \Phi_{xx}[0] = \frac{1}{2\pi}\int_{-\pi}^{\pi} \Phi_{xx}(e^{j\omega})\,d\omega$

(b) $\Phi_{xx}(e^{j\omega}) = \Phi_{ww}(e^{j\omega})\,\Phi_{hh}(e^{j\omega})$

$= \sigma_w^2 |H(e^{j\omega})|^2$

$= \frac{\sigma_w^2}{1.25 - \cos\omega}$

(c) $H(e^{j\omega}) = \frac{1}{1-0.5e^{-j\omega}} \Leftrightarrow h[n] = 0.5^n u[n]$

$\Phi_{xx}[n] = \Phi_{ww}[n] * h[n] * h[-n]$

$= \sigma_w^2 \delta[n] * h[n] * h[-n]$

$= \sigma_w^2\, h[n] * h[-n]$

$= \sigma_w^2 \sum_{k=-\infty}^{+\infty} h[k]\,h[k-n]$

$= \sigma_w^2 \sum_{k=0}^{+\infty} 0.5^k u[k]\, 0.5^{k-n} u[k-n]$

$= \sigma_w^2 \sum_{k=\max(0,n)}^{+\infty} 0.5^{2k-n}$

$= \frac{\sigma_w^2\, 0.5^{2\max(0,n)-n}}{1-0.25}$

$= \frac{\sigma_w^2\, 0.5^{|n|}}{0.75}$

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